

Project 3 Report - Semantic Duplicate & Near-Plagiarism Detection Engine

Objective: Compare MinHash+LSH and TF-IDF SimHash for near-duplicate document detection.

Dataset: Quora Question Pairs (Kaggle).

Preprocessing: normalization, tokenization, stopwords removal, 3-word shingles, TF-IDF weighting.

Results

Method	Accuracy	Precision	Recall	F1	Runtime
MinHash	0.645	0.545	0.083	0.144	60.1798 s
SimHash	0.635	0.478	0.152	0.231	60.0179 s

Discussion: SimHash obtained a higher Recall and F1 score while being significantly faster. MinHash achieved slightly better Precision but missed more duplicate pairs. Both methods achieved similar Accuracy (~64%).

Conclusion: SimHash provides the best balance between speed and effectiveness on the evaluated Quora subset.